



Figure 1: A layered approach to container technology
(Source: <https://www.Docker.com>)



Figure 2: A layered approach to Hypervisor and VM technology
(Source: <https://www.Docker.com>)

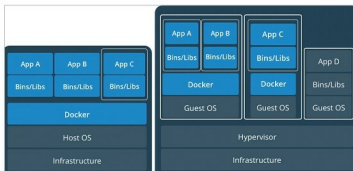


Figure 3: The hybrid approach of Docker containers and VMs
(Source: <https://www.Docker.com>)

DevOps, Cloud Foundry apps, virtual private networks and many others.

For the deployment and scaling of Docker applications on the IBM Bluemix cloud, a cloud based chat application called BlueChatter is used. This service uses Express, Node.js and RedisCloud.

Requirements to work with IBM Bluemix and containers include the following:

- An IBM Bluemix account
- An IBM Bluemix container service
- Cloning of the BlueChatter application with the local system
- Node.js
- Installation of Docker on the local machine
- Cloud Foundry CLI

To build and run the Docker container locally, first change the directory to the BlueChatter folder, which is cloned as follows:

```
$ cd bluechatter
```

Start building the Docker container, using the code below:

```
$ docker-compose build
$ docker-compose up
```

Now, the successful execution of the app can be checked from the URL <http://localhost> in case there is a Mac or a Linux operating system.

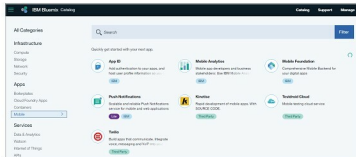


Figure 4: Catalogue of services in IBM Bluemix

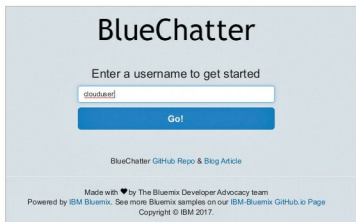


Figure 5: The IBM Bluemix based BlueChatter app



Figure 6: Chat environment in IBM Bluemix based BlueChatter app



Figure 7: Creating an IBM Bluemix account

For Windows, there is a need to map the IP address of the Docker machine for testing the application, as follows:

```
$ docker-machine IPAddress Name-of-the-Machine
```

Deploying the Docker image to IBM Bluemix

To begin, log in with the IBM Bluemix account with valid